
Lab 1

HW / SW Co-design of audio spectrum

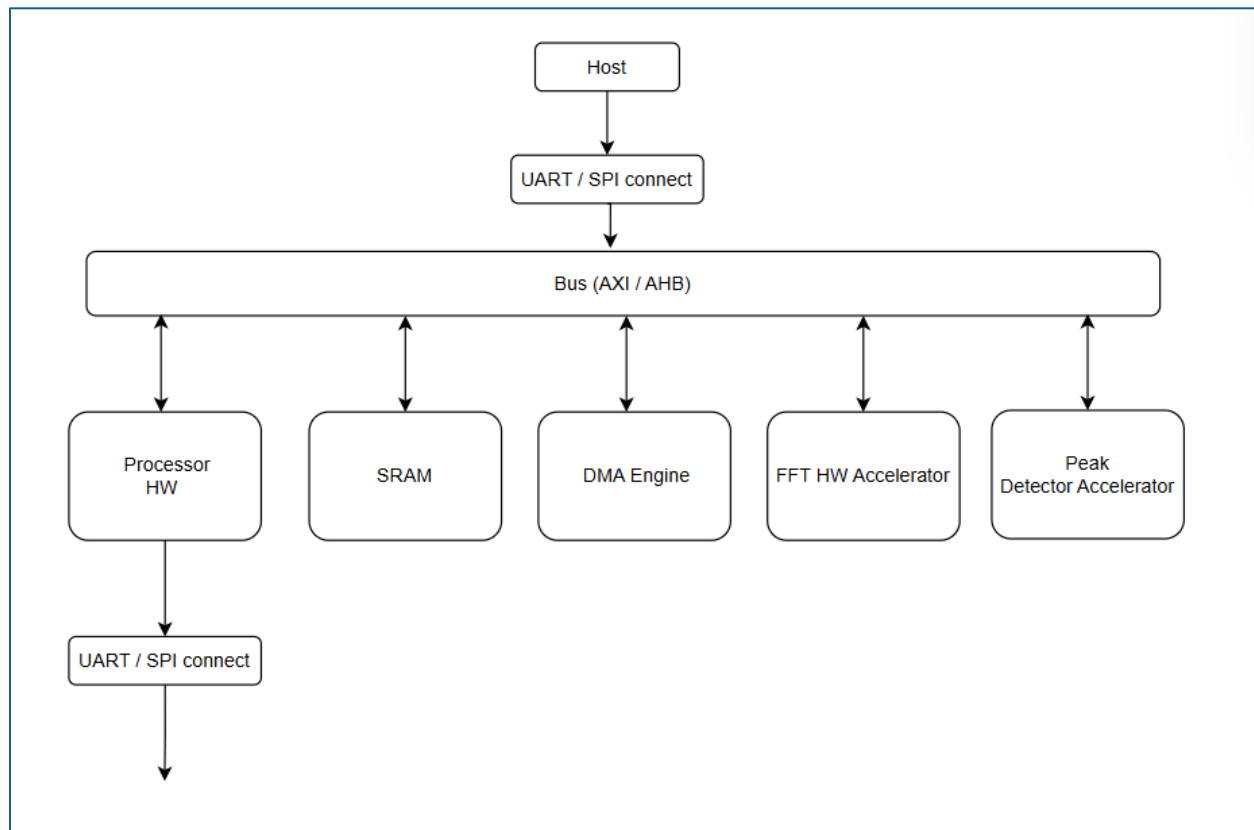
Submitted to:

Eng. Youssef Mohamed

Submitted by:

Menna Assem Fouda

1. SoC Arch.



2. HW / SW

Component	HW / SW	Reason
Processor (CPU)	HW	It is the hardware that executes the firmware and controls the whole system.
System Firmware / Application	SW	It controls the system flow and is easy to modify without changing the hardware.
UART/SPI Controller	HW	Serial communication requires accurate timing, so dedicated hardware is more efficient.
AXI/AHB Bus (Interconnect)	HW	It transfers data between all hardware blocks inside the SoC.
DMA Engine	HW	It transfers data between UART and SRAM without using the CPU, improving performance.
SRAM	HW	It stores the input audio samples and intermediate data.
FFT Accelerator	HW	FFT requires many arithmetic operations. Hardware performs them much faster using parallel processing.
Peak Detector Accelerator	HW	It quickly finds the maximum FFT magnitude using simple hardware comparators.
Frequency Calculation	SW	It is a simple equation, so there is no need for dedicated hardware.

System Initialization	SW	Configuration is easier to implement and modify in software.
System Control (Start FFT, Wait, Read Result, Send Result)	SW	These are control tasks that are simpler and more flexible in software.

3. Psuedo code

Start

Initialize UART/SPI

Initialize DMA

Initialize FFT Accelerator

Receive audio samples from Host

Store samples in SRAM

Start FFT Accelerator

Wait until FFT is finished

Find the peak frequency bin

Calculate the frequency value

Send the frequency back to the Host through UART/SPI

End